GNSS_Processing_fg Processing Pipeline

LIGO GNSS Pipeline Summary

2 August 2026

1 Purpose and scope

`GNSS_Processing_fg.cpp` connects filtered rover observations, ephemerides, LiDAR/IMU states, receiver-clock states, RINEX base-station carrier observations, and the GTSAM/iSAM2 graph. It first initializes the local-to-ECEF relationship and then incrementally adds measurement and motion factors, optimizes the float state, optionally fixes carrier ambiguities, and exports the optimized navigation state.

The primary entry points are `processGNSS()`, which validates and buffers observations, and `Evaluate()`, which constructs and optimizes the graph for an initialized epoch.

2 Pipeline overview

```
rover observations + ephemerides + LiDAR/IMU state
                         |
                         v
          processGNSSBase(): filter and associate
                         |
              +----------+-----------+
              |                      |
       not initialized         initialized
              |                      |
       fill time window      buffer current epoch
              |                      |
         GNSSLIAlign()            Evaluate()
              |                      |
       coarse ECEF position    initialize graph values
       and local/ECEF align          |
              |                AddFactor()
              |                      |
              |       +--------------+----------------+
              |       |       |       |       |        |
              |     P/D    clocks   motion  carrier   DD multi-frequency
              |       |       |       |       |        |
              +-------+-------+-------+-------+--------+
                                      |
                                iSAM2 float update
                                      |
                             ambiguity covariance + LAMBDA
                                  |                 |
                                reject            accept
                                  |                 |
```

```
keep float state   add integer priors
                            |
                   second iSAM2 update
                            |
                export optimized state
```

3 Inputs and graph state

The processor consumes rover pseudorange, Doppler, carrier phase, uncertainty, and LLI observations; broadcast ephemerides and ionosphere parameters; the current LiDAR/IMU state and preintegration result; and, optionally, a local RINEX 3.x base observation file, ZIP archive, or directory.

Key	Value	Role
A_k	position and velocity	Local LiDAR-enabled navigation state
R_k	rotation	Platform attitude
O_k	motion and biases	Angular rate, acceleration, IMU biases
G_k	gravity vector	Per-frame gravity state
F_k	ECEF navigation vector	Direct state without LiDAR
B_k	four clock biases	GPS/GLO/GAL/BDS receiver clocks
C_k	clock drift	Common receiver clock drift
E_0, P_0	translation and rotation	Local-to-ECEF alignment
n_j	scalar in metres	One reference/subject/band ambiguity arc

4 Observation intake and initialization

`processGNSS()` calls `GNSSAssignment::processGNSSBase()` to apply quality checks, associate ephemerides, and maintain Hatch-smoothed pseudorange. Before initialization, epochs with fewer than four valid satellites are rejected. Valid GNSS data and simultaneous local pose and velocity are stored in a `WINDOW_SIZE+1` initialization window.

`GNSSLIAlign()` requires a complete and temporally continuous window. Coarse single-point localization estimates ECEF position and constellation clock bias. With little local motion, the coarse ECEF point and an ENU rotation initialize the alignment. With sufficient motion, coarse ECEF positions across the window are aligned to the local trajectory. Failed localization, excessive time gaps, or failed alignment postpone initialization rather than seeding an invalid graph.

`SetInit()` inserts priors for navigation, alignment, clocks, drift, biases, and gravity. An initial iSAM2 update completes initialization.

5 Per-epoch factor construction

`Evaluate()` computes the epoch interval, selects the LiDAR-enabled or direct-ECEF representation, initializes the new graph values, and calls `AddFactor()`.

5.1 Pseudorange, Doppler, and clocks

For each satellite, transmit time, position, velocity, satellite clock, group delay, ionosphere, and measurement uncertainty are calculated. Pseudorange and Doppler constrain position, velocity, constellation clock bias, and common clock drift. Robust noise models are available. Clock evolution factors connect consecutive B_k and C_k states. This legacy path continues to select the established primary frequency.

5.2 LiDAR/IMU motion

The LiDAR-enabled branch normalizes the supplied information matrix and adds a `GnssLioFactor`. A non-positive diagonal marks the LiDAR contribution invalid instead of causing division by zero. The no-LiDAR branch uses IMU preintegration to connect consecutive direct-ECEF states.

5.3 Time-differenced rover carrier phase

The processor pairs the current primary-frequency carrier with an earlier continuous observation of the same satellite. The unchanged ambiguity cancels across time, constraining motion and clock change without an absolute ambiguity state. Histories are pruned by graph frame and elapsed time.

6 Base-station multi-frequency RTK path

6.1 RINEX 3.02 ingestion

`BaseStationData` accepts a direct RINEX observation file, a ZIP archive, or a directory containing multiple files or archives. It reads ZIP members without extraction and aggregates consecutive epochs. The surveyed antenna coordinate comes from `APPROX POSITION XYZ`; multiline `SYS / ## / OBS TYPES` records define each constellation's fields. Carrier values and LLI are decoded for GPS/Galileo L1 and L5, BeiDou B1I and B2a, and GLONASS L1. GLONASS channels come from `GLONASS SLOT / FRQ ##`. Special event records are skipped according to their declared record count.

The time system is normalized to GPST, and duplicate tracking codes at the same satellite/frequency are collapsed. Since SSI is signal strength rather than phase uncertainty, `base_carrier_std_cycle` supplies the assumed standard deviation. Multiple files must report a consistent ECEF antenna coordinate. The configured HKSC archives contain observations from 21 March 2025, irrespective of the enclosing directory's 2026 name.

6.2 Double-difference construction

The nearest base epoch must satisfy `base_epoch_tolerance`. Rover and base records are matched by satellite and physical frequency, then grouped by constellation and signal band. A reference satellite is selected separately for each group. With subject s and reference q, the carrier measurement is

$$\Delta\nabla L = (\lambda_s \Phi_r^s - \lambda_s \Phi_b^s) - (\lambda_q \Phi_r^q - \lambda_q \Phi_b^q), \tag{1}$$

and the geometric term is

$$\Delta\nabla\rho = (\rho_r^s - \rho_b^s) - (\rho_r^q - \rho_b^q). \tag{2}$$

The factor residual is $e_{DD} = \Delta\nabla\rho + n_j - \Delta\nabla L$. Local-frame and direct-ECEF variants provide analytical Jacobians.

Each ambiguity identity contains its reference satellite, subject satellite, and band. L1 and L5 therefore use independent wavelengths, reference selection, and ambiguity arcs. Loss of lock, a reference change, or a sample gap starts a new float ambiguity so an old integer constraint cannot leak into a new arc.

7 Float solution and LAMBDA fixing

The first iSAM2 update produces ambiguities in metres. Active ambiguities with sufficient lock duration are converted to cycles together with their full joint covariance:

$$\hat{a}_i = \frac{n_i}{\lambda_i}, \qquad Q_{a,ij} = \frac{Q_{n,ij}}{\lambda_i \lambda_j}. \tag{3}$$

LAMBDA solves

$$\check{\mathbf{a}} = \arg\min_{\mathbf{z}\in\mathbb{Z}^m} (\hat{\mathbf{a}} - \mathbf{z})^T Q_a^{-1} (\hat{\mathbf{a}} - \mathbf{z}). \tag{4}$$

Fixing requires the minimum ambiguity count and lock duration, the covariance precision gate, and the ratio test $s_2/\max(s_1, 10^{-12})$ above the configured threshold. Partial ambiguity resolution removes the highest-variance state and retries. Rejection leaves the float solution unchanged. Accepted integers are converted back to metres and inserted as tight fix-and-hold priors; a second iSAM2 update immediately propagates the constraints into the navigation state.

8 State export, fixed-lag management, and reset

Factor indices are recorded per frame. Old factors and variables are removed when the active window exceeds the deletion threshold, and carrier histories are pruned consistently. After optional integer fixing, `updateStateFromEstimate()` copies optimized attitude, position, velocity, biases, and gravity to the external state.

Missing base epochs, unavailable covariance, or failed integer validation disable only the optional fixed/DD update; the float GNSS/LiDAR/IMU graph can continue. `Reset()` clears all buffers, histories, ambiguity arcs, factor identifiers, initialization state, and fix status, and creates a fresh iSAM2 instance.

9 Important configuration parameters

Parameter	Default	Effect
`use_double_differences`	true	Enable base-station RTK factors
`use_secondary`	true	Add GPS L2/Galileo E5b/BeiDou B2
`use_l5`	true	Add supported L5/E5a/B2a carriers
`base_station_file`	HKSC directory	RINEX file, ZIP, or directory
`base_epoch_tolerance`	0.05 s	Rover/base synchronization gate
`rtk_ambiguity_gap_tolerance`	1.5 s	Continuity gate for 1 Hz base epochs
`base_carrier_std_cycles`	0.01 cycles	Assumed RINEX phase uncertainty
`double_difference_sigma`	0.02 m	DD factor standard deviation
`ambiguity_prior_sigma`	100 m	Initial float ambiguity prior
`enable_integer_fixing`	true	Enable LAMBDA and fix-and-hold
`lambda_min_ambiguities`	3	Dataset partial-fix subset
`lambda_min_lock_epochs`	10	Required continuous lock
`lambda_ratio_threshold`	3.0	Integer validation threshold
`lambda_max_std_cycles`	0.25	Float ambiguity precision gate
`fixed_ambiguity_sigma_cycles`	0.001	Integer-prior sigma
`rtk_debug`	true	Structured DD/float/LAMBDA logging
`rtk_debug_epoch_interval`	10	Repeated diagnostic period in epochs

10 Interpreting RTK diagnostics (2 August 2026)

The runtime separates three milestones. `[RTK-DD]` confirms that a base epoch was synchronized and that DD carrier factors entered the graph; explicit missing-base and zero-factor counters diagnose failures before optimization. `[RTK-FLOAT]` confirms that ambiguity states exist and reports their cycle estimates, standard deviations, and lock eligibility. `[RTK-LAMBDA]` reports every validation gate. Only a line containing `FIX SUCCESS` means an integer RTK solution was accepted and tight fix-and-hold priors were added. A `NO FIX` or `FLOAT_REJECTED_*` status is still a floating-point solution, with the specific failed gate shown in the same log stream.

The 2 August replay verified continuous key reuse across 1 Hz base epochs, correct handling of Galileo LLI bit 2, stable reference selection, and actual entry into LAMBDA after ten locked epochs. Three eligible float ambiguities had standard deviations of roughly 0.85–1.41 cycles, so the conservative 0.25-cycle precision gate correctly retained the float solution.

11 Current multi-frequency boundary

Primary, GPS L2/Galileo E5b/BeiDou B2, and L5 are used in the synchronized base/rover DD path and joint LAMBDA resolution. The existing pseudorange/Doppler and rover-only time-differenced paths remain on their primary frequency. Raw per-frequency integer ambiguities are kept; no ionosphere-free combination is formed because its combined ambiguity is not a direct integer in cycles. GLONASS remains L1-only in this mapping.

12 Verification status

The Release executable builds and links successfully. Eight RTK core tests verify both DD factor variants and all analytical Jacobians, multi-frequency selection, LLI masking, RTK arc continuity, LAMBDA against exhaustive integer search, translation invariance, and invalid inputs. Two base-reader tests verify a compact dual-frequency RINEX 3.02 fixture and direct loading of both actual HKSC ZIP archives with consecutive-hour aggregation. Further diagnostics print GPS, GLONASS, Galileo, and BeiDou frequency counts and scan the rover bag. The synchronized inputs yield 304 matched epochs, 4449 matched carriers, and 3011 potential DD factors. The latest result is fifteen passing LIGO test cases; the catkin collector reports 30 tests, zero errors, zero failures, and zero skipped tests.

The real rover bag contains no L5 observations. Its LiDAR topic also uses `livox_ros_driver2/CustomMsg`, while this executable expects `livox_ros_driver/CustomMsg`; therefore a driver adapter remains necessary before a complete live factor-graph replay can evaluate positioning accuracy and ambiguity-fix convergence.